Matthew **Grinaker**

AI Product Engineering Lead | CTO | Agentic Systems

E: matt@g-techsystems.com
P: +27 83 659 1690
L: www.linkedin.com/in/mk-grinaker

KZN, South Africa
EU Citizenship

About

Software engineering leader with 10+ years designing, building and running production software across enterprise SaaS, automation, trade finance and mission-critical industrial systems. Strong commercial background in C#/.NET, Azure, multi-tenant platforms and leading engineering teams.

Since 2025 I've designed and built **AI agent systems** with the Claude Agent SDK, MCP, governed tool calling, multi-agent orchestration and evaluation-driven release pipelines. The work covers AI-assisted software delivery, controlled access to enterprise systems, and document intelligence with source attribution and human feedback.

I stay hands-on in architecture, code and review while owning technical direction, delivery standards and product outcomes. My focus is AI that is useful, reliable and accountable in real business operations. I give AI more responsibility only as evals show it can carry it, and an old shortcut never becomes the new standard.

Core Leadership & Strategy

Technical Leadership · Team Management · Agile & Scrum · Stakeholder Management · Product Roadmaps · AI Engineering Standards · Quality Assurance

Applied AI & Agent Engineering

Multi-Agent Orchestration · MCP Servers · Agent Evals & Graders · Tool Permissions & Audit · Context Engineering · Progressive AI Autonomy

Software Engineering & Architecture

System Design · Event-Driven Architecture · Multi-Tenant SaaS · API-First Design · Microservices · Modular Codebases

Technical Proficiencies

Programming Languages: C# · Python · TypeScript · JavaScript · SQL · React · Tailwind CSS · Angular · Delphi · C++ · Bash/Shell

AI & Agents · Claude Agent SDK · Claude Code (headless) · Anthropic API · MCP (FastMCP, SDK servers) · Tool Calling · Evals & LLM Graders · Hooks, Skills & Plugins · Prompt & Context Engineering · OpenAI & Google AI APIs · GitHub Copilot

Backend & Messaging · .NET Core / ASP.NET Core · Entity Framework Core · Node.js / Express · FastAPI · RESTful APIs · Redis / Bull Queues · MSMQ · SignalR

Cloud & DevOps · Azure App Services · Azure SQL · Key Vault · Azure AD / RBAC · Application Insights · Azure DevOps · Docker · GitHub Actions · GitLab CI

Databases · PostgreSQL (RLS) · MSSQL · MySQL · SQLite · Firebird

Networking · OpenWRT · TCP/IP client-server protocols · WinSock · CISCO CCNA & CCNP

Education

2004 - 2009
Hilton College
IEB Matric

2010 - 2014
University of Pretoria
B.Eng Computer Engineering

Professional Experience

2021 - Present

Grinaker Technologies

Technical Consultant

Duties & Responsibilities

- **Product:** Set the technical direction and roadmap for client and in-house products, and sit with business users to map how the work actually runs (trade finance, ERP, logistics, property, industrial automation).
- **Architecture:** Design multi-tenant SaaS on PostgreSQL (schema per tenant, row-level security) with job queues and ERP integrations.
- **AI in the product:** Build AI features customers use, such as StockChain's document reader, and decide where AI should help, act or stay out.
- **AI for delivery:** Run our engineering through AI agents on Claude Code and Codex, with MCP tools, evals, an independent grader on every ticket and my approval on every merge.
- **Team:** Mentor engineers and set our coding standards, reviews and release gates.

Achievements

- Shipped StockChain to its first live customer in 7 months: ~260k lines of backend TypeScript, 36 modules, 490+ migrations, 1,100+ test files.
- Built an agent orchestrator that has delivered 130+ PRDs in parallel, each checked by an independent grader.
- Built a multi-tenant deal system on ASP.NET Core: 200+ API endpoints, 50+ entities, AES-256 encryption on all personal data.
- Ran real-time IoT systems across 1,000+ devices.
- Grew users 40% and delivered 90% of projects on time.

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Professional Experience (continued)

2024 - 2025

Prosolutions Integration

Senior Software Developer & Team Lead (contract, alongside Grinaker Technologies)

Duties & Responsibilities

- Wrote and reviewed production code for client integration projects.
- Led the development team and mentored developers.
- Planned and ran projects with clients, from requirements to release.
- Owned code review and QA before anything shipped.
- Found and fixed performance problems in client systems.
- Acted as the technical point of contact for clients.

2019 - 2021

GoMobile (Pty) Ltd

Chief Technical Officer · C#/.NET on Azure

Duties & Responsibilities

- Owned the architecture and tech stack of Magnitude, a multi-tenant no-code platform used by 30+ client organisations.
- Ran a 12-person Agile team (10 developers, 2 testers): sprint planning, code reviews and one-on-one coding sessions.
- Ran our Azure estate: App Services, Azure SQL, Key Vault, Azure AD and Application Insights.
- Built CI/CD in Azure DevOps with automated test gates (NUnit, xUnit).
- Led POPI compliance: encryption, access control, pen tests, audit trails and data retention.
- Dealt with clients, dealers and stakeholders on security and compliance.

Achievements

- Rebuilt the backend with zero downtime: p95 API response went from 2-5 s to under 500 ms.
- Tripled concurrent users (50 to 150+) with database sharding and Redis caching.
- Cut production incidents by 60%, from 2-3 a month to under 1.
- Released 30% more often (every 3 weeks to every 2).
- Passed 100% of third-party POPI security audits.

Stack: C#, .NET Core, ASP.NET Web API, EF Core, Azure SQL, Redis, Angular, Azure DevOps

2016 - 2019

Nerospec

Senior Software Engineer (2017-2019)
Software Engineer (2016-2017)

Duties & Responsibilities

- Designed GPS route control and collision avoidance for rail vehicles in working mines.
- Built the control-room software for route planning, with a custom vector map of live positions.
- Wrote multithreaded C# and C++ Windows services that collect telemetry over TCP/IP and UDP.
- Built a monitoring and control tool for industrial networks (SNMP, SSH, SMS, JSON-RPC, MikroTik).
- Designed MSSQL storage for telemetry, with SQLite caching on edge devices.

Achievements

- Processed 500-1,000 telemetry points a second with sub-second latency.
- Zero safety incidents across 24/7 mine operations.
- Kept mission-critical services at 99.9% uptime.
- Cut manual network intervention by 60%+.

Stack: C#, C++, Delphi, .NET Framework, MSSQL, SQLite, WinSock, Grafana

AI Engineering & Agentic Systems

What I've built and use every day at Grinaker Technologies, 2025 to now.

Agent delivery pipeline

Splits each PRD into dependent tickets and runs every ticket as a headless Claude Code worker in its own git worktree, with fixed limits on model, turns and budget. Work only lands on an integration branch I merge. A final audit turns gaps into new tickets, up to 3 rounds.

Evals and quality gates

Nothing merges without a pass. A grader with no shared context re-runs the checks and returns a JSON verdict. A validator agent scores 7 weighted criteria with 10 hard gates. Accuracy baselines for each document type fail CI if they drop.

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MCP servers and tool permissions

I wrote two read-only MCP servers over live business data: Odoo ERP (9 tools, method allow-list, works on Odoo 14 to 19) and tenant-scoped PostgreSQL (read-only transactions, one SELECT per call). Agents get only the tools they need, a spend cap per run, and approvals that default to no.

Business context for agents

164 versioned architecture decisions, each with its source and linked to the code it covers. A hook blocks an agent's edit until it has read the relevant ones, even after context compaction. A shared second brain adds the team's decisions, handovers and live work claims, so every session starts from the same picture.

Agent SDK runtime

A conductor agent plans the work and starts researcher, coder, reviewer and tester agents on a replayable message bus. Hard caps on budget, depth and concurrency, loop detection, human approval gates, and a prompt-injection scan on everything it reads.

Usage, cost and verdict tracking

A monitor reads Claude Code and Agent SDK transcripts and splits tokens and cost by model and session, with a daily report. Hook events stream to a local service that scores grader verdicts against their rubrics.

Claude Code extensions I wrote: 6 skills (delivery orchestrator, PRD generator, ERP and freight API clients), 16+ agents mirrored for Codex, 2 plugins that expose MCP tools, and guard hooks for secrets, bash bypass, typechecks and stop-time validation.

Recent Projects

2026

StockChain · Data platform for import merchants

Live with its first customer after 7 months. One deal flow, from RFQ to payment, coordinates the merchant, customer and lender with ERPs, freight agents and FX brokers. The document reader learns layouts from what users confirm, and every value keeps its source and confidence, with accuracy baselines checked in CI.

Tech: TypeScript, Node/Express, PostgreSQL (RLS), Redis/Bull, React 19, Docker

Features: ERP sync with reconciliation, per-tenant data isolation (RLS), OCR

Highlights: ~2,700 commits, 1,100+ test files, built with my agent pipeline

Shared Second Brain · Shared context and tooling for a team's AI agents

In use by the StockChain team since October 2026, built after two AI agents undid each other's settled work. One shared place holds the business data, context and rules every agent works from, plus the team's agents, skills, hooks, mods and tools. Each session starts from the same picture, and an agent has to claim an area before it can change it.

Tech: Node.js CLI (no dependencies), GitHub Issues and git as the store, Claude Code hooks

Features: Decision log with supersession, handovers, work claims with expiry, shared agents, skills and mods

Highlights: Owner, source and last-verified date on every record; design passed independent grading

Claude Fleet · Control room for AI agents

I see and steer every Claude Code session on my Mac and in the cloud from one page, and answer agents from Slack when I'm away.

Tech: Node.js, Electron, Slack Bolt, Claude Code hooks

Features: Live task lists, nested subagents, message any session

Highlights: Every write needs approval, self-update with rollback

GTech OS · Multi-agent platform

On its first live run, three agents planned, built and reviewed a task in two attempts. The LLM makes the decisions; plain code enforces the rules.

Tech: TypeScript, Claude Agent SDK, JSONL event bus

Features: Budget ledger, approval gates, restart recovery

Highlights: Loop detection, injection scanning, full replay

2025

i-Con Web Monitor

Cut the time to spot a failed integration server from 30+ minutes to under 2, across 100+ live connections.

Tech: C#, .NET 8, ASP.NET Core, SignalR, React, Redis

Features: Custom TCP protocol client, live dashboard

Highlights: 10,000+ messages a minute, sub-100 ms API